

Loan Approval Prediction – Model Evaluation Report

1. Project Objective

The objective of this project was to build a supervised machine learning model to predict whether a loan application would be approved based on applicant information such as income, loan amount, credit history, education, employment status, and property area.

The project focused on:

- Data preprocessing
 - Handling class imbalance
 - Model comparison
 - Performance evaluation
 - Business-oriented interpretation
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2. Data Preprocessing

Several preprocessing techniques were applied before model training:

Missing Value Handling

Missing values in both numerical and categorical features were imputed using median and mode values respectively.

Feature Engineering

Additional features were created to improve model performance:

- $\text{TotalIncome} = \text{ApplicantIncome} + \text{CoapplicantIncome}$
- ApplicantIncome_Log
- LoanAmount_Log

Log transformations were applied to reduce skewness and improve the representation of income and loan amount distributions.

Encoding

Categorical variables were converted into numerical values using label encoding.

Scaling

StandardScaler was applied before training Logistic Regression to ensure all features were on a comparable scale.

3. Class Imbalance Handling

The dataset contained more approved loans than rejected loans, resulting in class imbalance.

SMOTE (Synthetic Minority Oversampling Technique) was used on the training data to generate synthetic samples of the minority class and create a balanced dataset.

This helped improve the model's ability to identify rejected loan applications.

4. Model Comparison

Two classification models were trained and evaluated.

Logistic Regression

- Precision: 0.874
- Recall: 0.894
- F1 Score: 0.884
- ROC-AUC: 0.793

Random Forest

- Precision: 0.851
 - Recall: 0.871
 - F1 Score: 0.860
 - ROC-AUC: 0.818
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5. Model Trade-Off Analysis

Logistic Regression

Advantages:

- Higher Precision
- Higher Recall
- Higher F1 Score
- Easy to interpret
- Suitable for explaining approval decisions

Disadvantages:

- Slightly lower ROC-AUC score
- May not capture complex nonlinear relationships

Random Forest

Advantages:

- Highest ROC-AUC score
- Captures nonlinear patterns
- Provides feature importance information

Disadvantages:

- Less interpretable than Logistic Regression
- Slightly lower Precision, Recall, and F1 Score

Selected Model

Although Random Forest achieved the highest ROC-AUC score, Logistic Regression demonstrated better Precision, Recall, and F1 Score.

Since loan approval systems require reliable classification performance and explainability, Logistic Regression is recommended as the primary deployment model.

6. Business Interpretation

Feature importance analysis showed that the following variables had the strongest influence on loan approval decisions:

1. Credit_History
2. ApplicantIncome
3. TotalIncome
4. ApplicantIncome_Log
5. LoanAmount_Log

Key Insights:

- Applicants with a positive credit history have a significantly higher chance of loan approval.
- Higher income levels increase approval probability.
- Loan amount and repayment capability strongly affect approval decisions.
- Demographic attributes such as gender and self-employment status have relatively low impact.

These findings align with real-world lending practices where creditworthiness and repayment capacity are the most important factors.

7. Suggested Deployment Threshold

The default classification threshold is 0.50.

For loan approval systems, rejecting a genuinely eligible applicant can negatively impact business growth and customer acquisition.

Therefore, a slightly lower threshold is recommended:

Recommended Threshold = 0.40

Benefits:

- Increases recall
- Reduces false rejections of eligible applicants
- Improves customer approval coverage
- Maintains acceptable precision levels

The final threshold should be selected based on the organization's risk tolerance and lending policy.

8. Conclusion

The project successfully developed and evaluated machine learning models for loan approval prediction.

Both Logistic Regression and Random Forest produced strong results, with Logistic Regression achieving the best balance between Precision, Recall, and F1 Score, while Random Forest achieved the highest ROC-AUC score.

The analysis demonstrated that credit history and income-related features are the most influential factors in loan approval decisions. Based on overall performance and interpretability, Logistic Regression is recommended for deployment with a threshold of 0.40.

GitHub Link: https://github.com/Afzal-gif888/Loan_Approval_Predictor

NoteBook Link: https://github.com/Afzal-gif888/Loan_Approval_Predictor/blob/main/Loan_Approval_Prediction.ipynb